



OPERATORS FOR SWING GATES

GR 300 - GR 400 - GR 600 / 220V A1

INSTALLATION MANUAL

Our compliments for your excellent choice. The GR300-400-600 electro-mechanical gear motor has been produced for reliability and high quality.

This Manual will offer information you may need to install your gear motor assuring long-lasting performance and to safeguard your safety.

HOWEVER CAUTION IS UNQUESTIONABLY INDISPENSABLE AND NOTHING IS BETTER THAN PREVENTING ACCIDENTS.

GR products have been made to conform with rules and laws in force at time of manufacture.

! This manual is designed exclusively for the specialized installation expert in the criteria of construction and equipment to assist in the protection against accidents in the installation and use of the gate; door and automation of such gates (adhere to the rules and laws in force).

! On completion the installer should issue to the end consumer an instruction manual according to EN 12635.

! Before proceeding with the installation the installer must provide an analysis of the identification and management of risks as per the standards EN 12453 and EN 12445.

! All wiring of the various external electrical components connected to the automation (e.g. Photocells, flashing lights, keypads etc) must be carried out according to EN 60204-1 and the amendments made of the point 5.2.2 of EN 12453.

! It is prohibited to do any repair or adjustment of the equipment if you have not taken all necessary precautions to avoid possible accidents (example: power supply disconnected, engine block). All mechanisms in motion must be equipped with appropriate protections.

! The mains power line must be protected for maximum current in locked rotor condition as per government electrical laws.

! Install the gear motor on gates that conform to EN 12604.

! Perform the measure of strength developed by the gear motor and take the appropriate steps as per EN 12445.

! Positioning photocells: These safety devices must be installed at a height not exceeding 70cm from the ground and at a distance from the floor movement of the door of no more than 20cm. Their proper functioning of the photocells must be verified at the end of installation according to Section 7.2.1 of EN12445.

! Keep the activation controls of automation out of reach of children. The controls should be installed at a minimum 1.5m height above the ground and outside the range of actions of moving parts such as the gate.

! All activation actions must be executed only at points from where the automation is fully visible.

! Operate the remote only in view of automation.

! Store carefully this manual in a suitable place known to all interested people.

! Any unauthorized and arbitrary modification made to this product, releases the company GR Sistemi Automatici di apertura Srl and from any liability resulting from damage or injury to things, people or animals.

! The non-observance of regulations and of safety standards here listed releases the company GR Sistemi Automatici di apertura Srl from any liability resulting from damage or injury to things, people or animals.

! The automation must be coupled to a control board equipped with torque regulation that provides an anti crushing safety as described in EN 12453 - EN 12445

SAFETY RULES

During the installation and the use of the automation, pay attention to the following safety rules:

! Distance security!

! Mechanisms moving!

! Do not install automation in an environment saturated with explosive mixtures!

! Electric Shock!

! Use gloves!

! Use welding glasses!

! Maintain ear protection!

MAINTENANCE

! All repairs must be carried out by qualified people.

! Before each intervention remove power through the switch and lock in that position

! The equipment must be maintained so as to preserve the conditions that ensure safe and efficient operation

! Always use original spare parts

! Do not make interventions that modify the machine.

! The modified equipment requires new CE mark

! The settings of the operator must be performed by qualified personnel, in accordance with the rules of reference. During these operations provide the presence of two operators for safety.

DEMOLITION

You have to operate the elimination of the materials in conformity with the regulations in force. All materials must be divided by type (copper, aluminium, plastic, electrical parts, etc)

DISMANTLING

In order to move away the automation, follow these instructions:

- 1 - cut off the power supply and disconnect the electrical installation;
- 2 - dismantle the control console and all the other components of the installation.

If you have noticed that some components have been damaged, you have to replace them.

CONFORMITY DECLARATION:

It's in accordance with Machine Directive 39/89/CE and following modify.

It's in accordance with the following directive CE:

Electromagnetic compatibility Directive 89/336/CEE and following modify.

Low tension Directive 73/23/CEE and following modify.

Have been applied the following harmonized norms:

EN292/1/2, EN 294, EN60335-1, UNI EN 12453, and what applicable of the EN12445-2000.

USE OF THE AUTOMATION

The gearmotor GR300-400-600 was designed and built for the opening of gates with max of 5 m leaf or weight max. 250kg. The GR Srl assumes no responsibility for a purpose other than that provided by gearmotor GR300-400-600. Since automation can be put into motion in view by button or remotely by remote control, it is essential to check frequently the perfect efficiency of all safety devices. It is advisable to check periodically (every six months) the regulation of electronic friction of which must be equipped the electronic control board.

PRELIMINARY CHECKS

- 1 - Read carefully the instructions enclosed in this manual.
- 2 - Make sure that the gate has a rather solid structure and that there is no friction points in its movement.
- 3 - Make sure that the leaf is suitably balanced, even after the installation of the gear motor.
- 4 - Check that the electrical installation is in accordance with the characteristic required by the gear motor.

TECHINICAL DATA

	GR 300 A1	GR 400 A1	GR 600 A1
Maximum length of the gate	3,0m	4,0 m	5,0 m
Maximum weight of the gate	300Kg	300Kg	300Kg
Stroche	30cm	40cm	60cm
Power	350 W	350 W	350 W
Power supply	220 V	220 V	220V
Absorption	1,5 A	1,5 A	1,5 A
Max. Thrust	2200 N	2200 N	2200 N
Operating temperature	-25°C +60°C	-25°C +60°C	-25°C +60°C
Opening time	17sec.	22sec.	33sec.
Thermal protection	150°C	150°C	150°C
Use frequency	35%	35%	35%
Lubrication	GREASE		

SCHEDULED MAINTENANCE

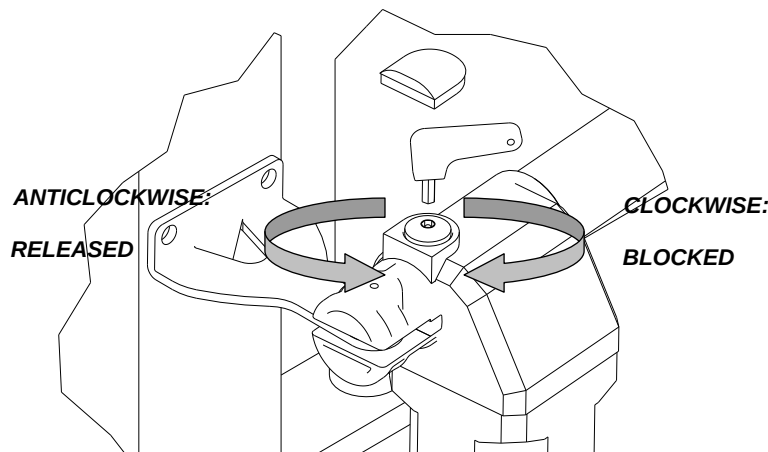
DESCRIPTION	FREQUENCY	ENTRUSTED	OPERATION
Photocells cleaning	Monthly	Operator	Clean with damp cloth
Control of gate hinges and supports, balancing of the gate	Annual	Operator	Check the status of welds and corrosion. Unhook the engine and check the balancing and the eventual points of friction.
Controlling the sensitivity of electronic friction (torque adjustment) of the control board.	Semiannual	Technician	Check the adjustment of the couple as described in EN 12453 - EN 12445
Monitoring current dispersion	Annual	Technician	Verify that the dispersion of current is less than 7.5 A
Control of signals	Semiannual	Operator	Verify that the safety warning signage is complete and intact

EMERGENCY MANOEUVRE

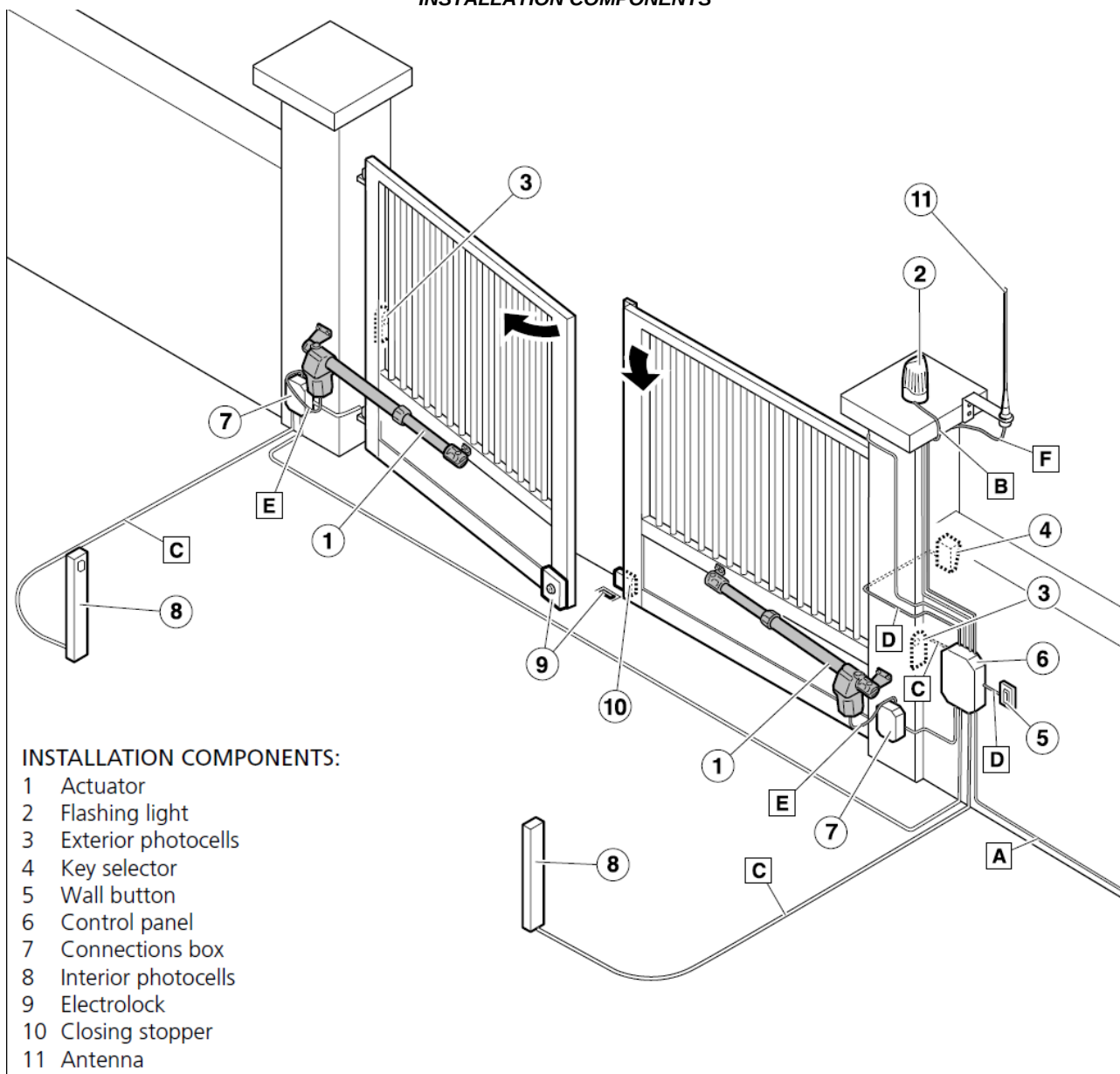
For the manual door locking and unlocking act with the supplied key on the screw C (See FIG 1-2).

- 1 - Remove the protection cap
- 2 - insert the key into the seat allocation as in Figure 1
- 3 - Turn the key in the sense of the arrow to the top of the gearmotor to unlock and in the opposite block.

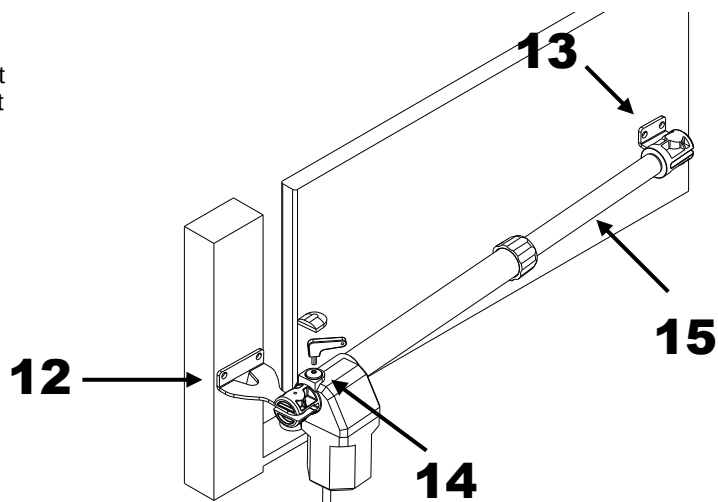
FIG 1



INSTALLATION COMPONENTS



- 12 – Pillar bracket
- 13 – Gate bracket
- 14 – Release
- 15 – Piston rod



TOOLS



SCREW DRIVERS



ALLEN KEYS



LEVEL



ELECTRIC DRILL AND BROACHES



ALLEN KEYS



PENCIL



SCREWS,NUTS, WHASHERS



FIXING ACCESSORIES



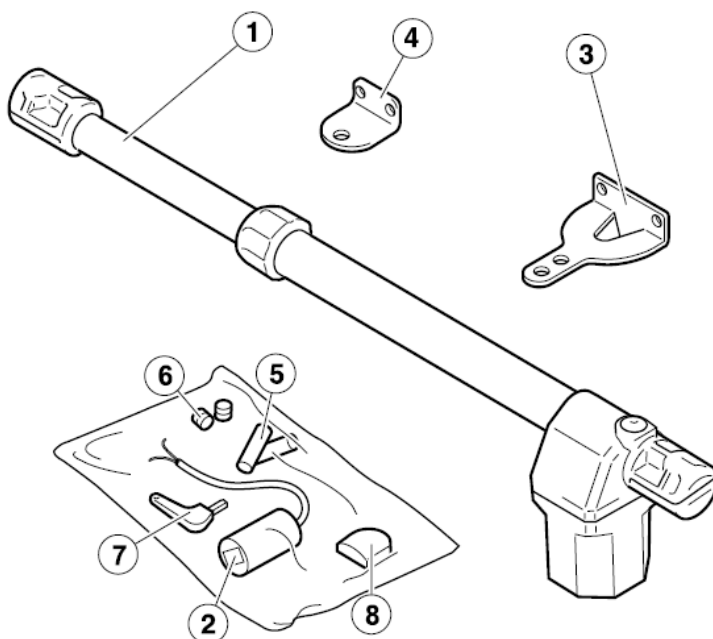
WELDING MACHINE



SAFETY EQUIPMENT



TAPE MEASURE



▲ Discard the packaging safely and in an environmentally friendly manner.

■ Check the content of the packages. Should it be noticed that a piece is missing or deteriorated, contact the nearest technical service.

- 1 Actuator
- 2 Capacitor
- 3 Wall support
- 4 Gate support
- 5 Shafts
- 6 Shaft caps
- 7 Unlocking key
- 8 Unlocking orifice cap

INSTALLATION

To operate a good installation of the gear motors GR 300/400/600, follow these instructions:

1 - Check that the size of the gate is within the admissible range of the actuator (see the technical characteristics of the actuator).

! If the gate to be automated has a passage gate, use a safety device to prevent the actuator from operating with the passage gate open.

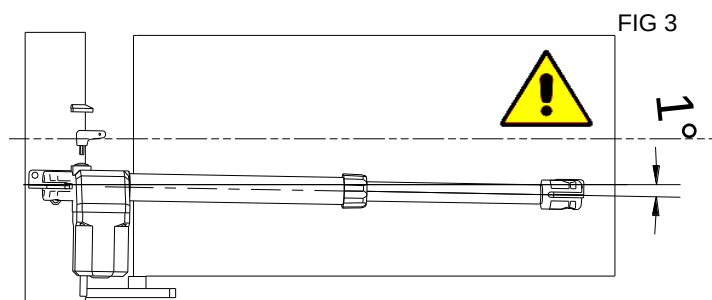
! The gate must be easy to manipulate manually, namely:

- It must be balanced, in order to ensure the effort made by the motor is minimum.
- There should be no stiffness throughout its travel.

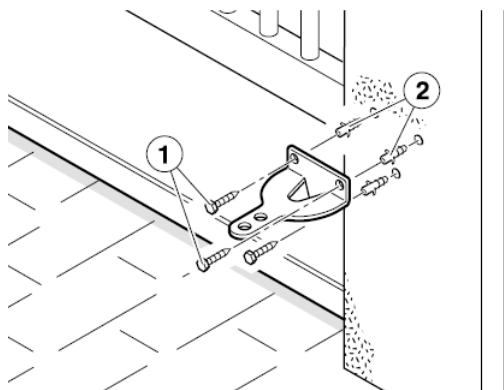
! Do not install the actuator in a gate which does not work correctly in manual operation, as this may lead to accidents. Repair the gate before installing.

2 - Make sure that the leaf of the gate is perfectly horizontal.

3 - Place the gear motor inclined approximately 1° below the horizontal line (picture 3)



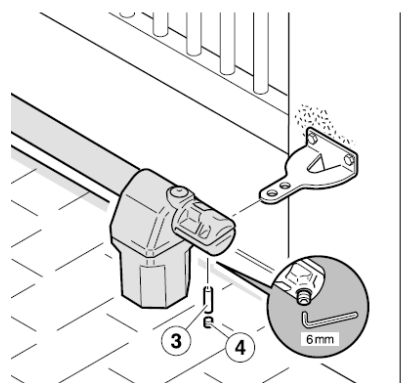
4- Fix the support plate 12 on the pillar beside the leaf, taking in account the measures shown in Table 1. Do not forget inclination.



! Respect the levels indicated in "Assembly positions and levels" table on next page

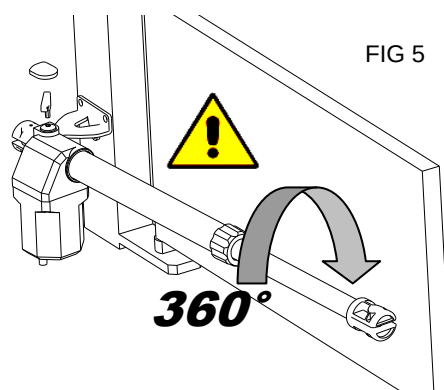
! The screws (1) and plugs (2) are not supplied. Use screws and plugs which are suitable for the place of attachment and the weight and effort of the actuator.
The bracket could be fixed also using a welding machine if required

5 - Mount the actuator on the wall support. Introduce the pin (3) and secure with the grub screw (4), using a 6 mm Allen key.

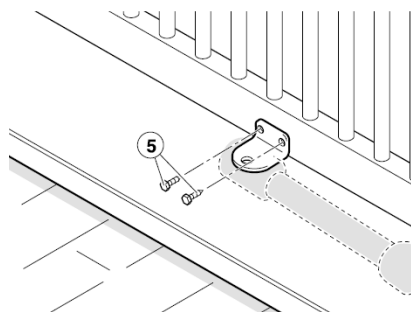


6 - With gate's leaf closed, turn and slide the screw of gear motor's 15 shaft, until the end of the stroke-

7 - Screw 15 shaft back on 1 complete turn of 360°. (picture5)



8 - Place B support plate in the hole of 15 shaft and position it against the gate leaf. Fix it to the gate leaf taking in account the inclination (point nr.3). The bracket could be fixed also using a welding machine if required

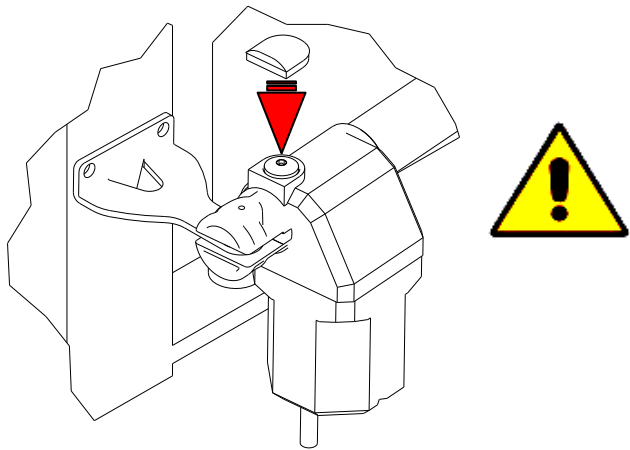


9 - Proceed in the same way with the other gate leaf.

10 - Connect the electrical wires and safety devices: Place the mechanical limit stops

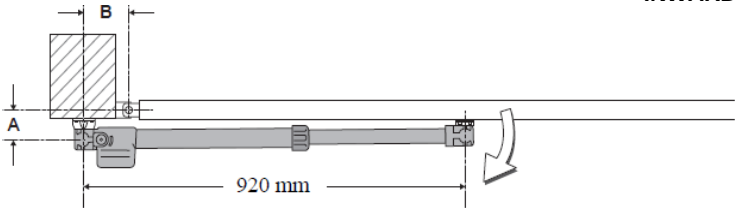
Place the cover on the unblocking screw (14) (picture 6)

FIG 6



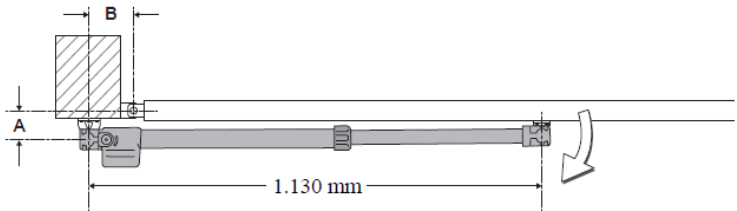
ASSEMBLY POSITIONS AND LEVELS
Levels A and B determine the opening angle C.

INWARD OPENING



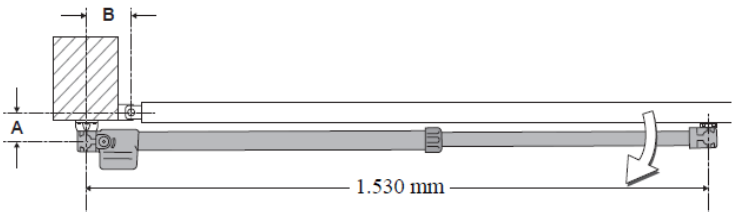
GR300

A(cm)	B(cm)	C(°)
140	140	95
120	160	120



GR400

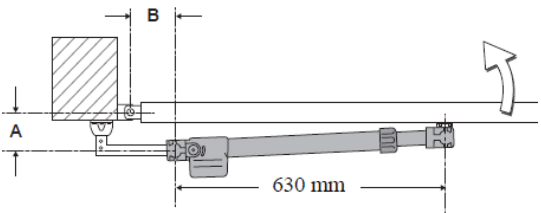
A(cm)	B(cm)	C(°)
145	145	95
110	170	120



GR600

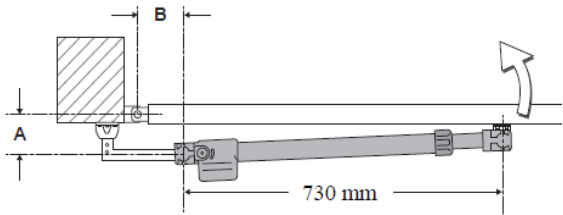
A(cm)	B(cm)	C(°)
280	280	95
120	310	120

OUTWARD OPENING



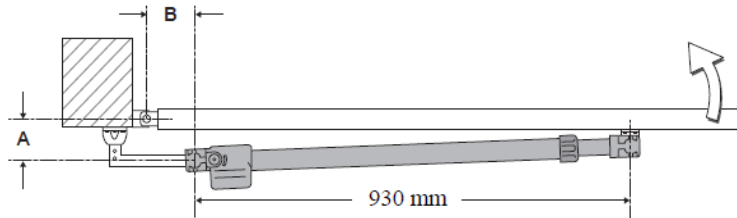
GR300

A(cm)	B(cm)	C(°)
140	140	95
160	120	120



GR400

A(cm)	B(cm)	C(°)
145	145	95
170	110	120

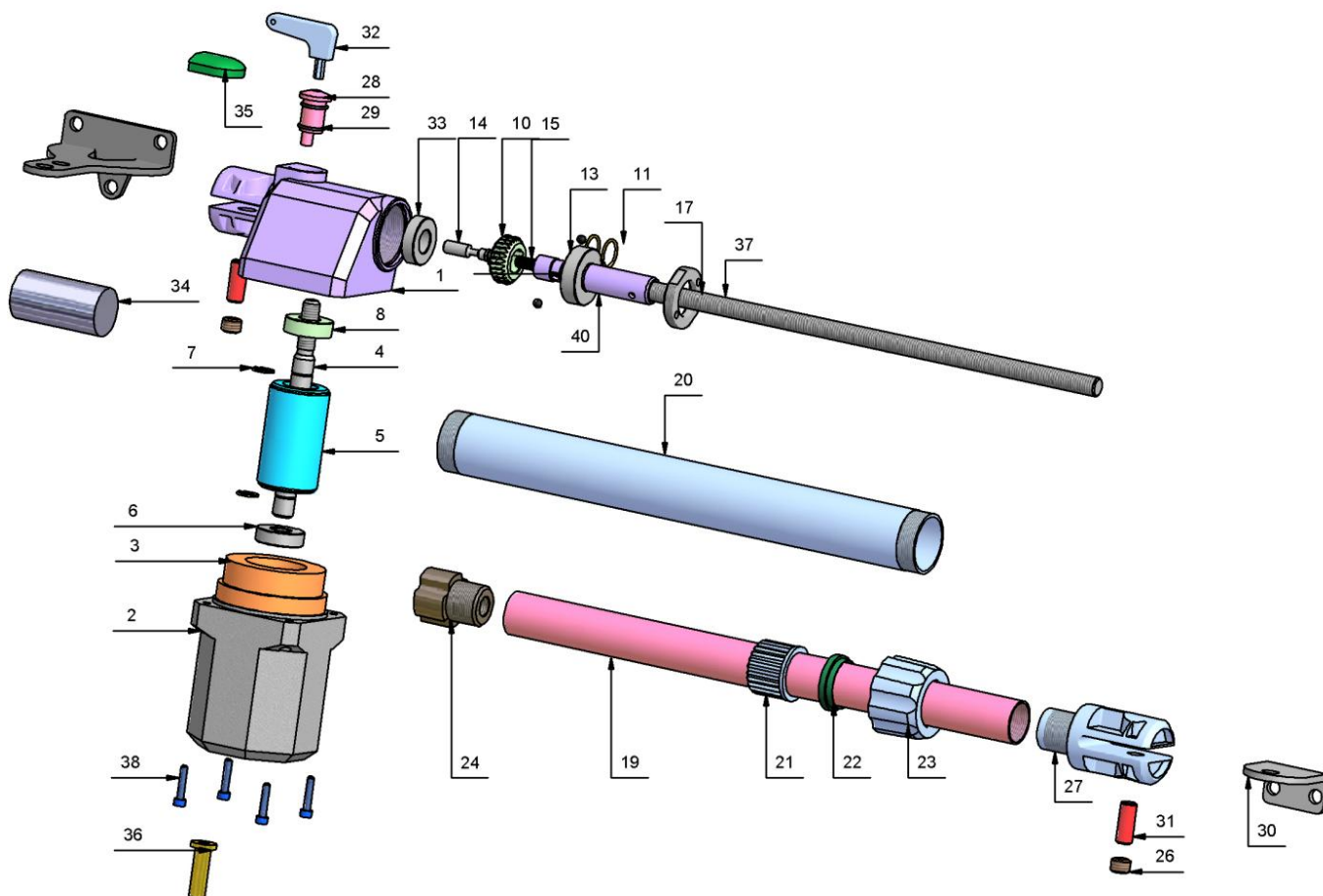


GR600

A(cm)	B(cm)	C(°)
280	280	95
120	310	120

SPARE PARTS

PART	QTY	CODE	DESCRIPTION	PART	Q.TY	CODE	DESCRIPTION
1	1	100645	REDUCTION GEAR COVER LF. BLACK	20	1	100715	BLACK ALUMINIUM TUBE 400mm STROKE
1	1	100644	REDUCTION GEAR COVER RG. BLACK	20	1	100763	GREY ALUMINIUM TUBE 400mm STROKE
1	1	100759	REDUCTION GEAR COVER LF. GREY	20	1	100739	BLACK ALUMINIUM TUBE 600mm STROKE
1	1	100758	REDUCTION GEAR COVER RG. GREY	20	1	100764	GREY ALUMINIUM TUBE 600mm STROKE
2	1	100647	MOTOR CASING RG. BLACK	21	1	100829	PISTON ROD GUIDE
2	1	100761	MOTOR CASING RG. GREY	22	1	100625	SCRAPER RING
3	1	100642	STATOR 220V 4 POLES 70mm 1400rpm	23	1	100827	PISTON ROD LOCK NUT
4	1	100651	IRREVERSIBLE MOTOR SHAFT	24	1	100828	FEMALE SCREW
5	1	100643	ROTOR 70mm	25	1	100823	WALL BRACKET
6	1	100278	BEARING 6202 ZZ	26	2	100705	GRUB SCREW M14X10
7	2	100706	HELASTIC RING E15	27	1	100646	ALUMINIUM PISTON ROD TERMINAL BLACK
8	1	100291	BEARING 6202 2RS	27	1	100906	ALUMINIUM PISTON ROD TERMINAL GREY
10	1	100650	IRREVERSIBLE TOOTH WHEEL	28	1	100624	ALUMINIUM UNCLAMPING
11	2	100789	HELASTIC RING E20	29	2	100657	O-RING
12	1	100709	BEARING 6004 ZZ	30	1	100825	GATE BRACKET
13	2	100658	TOWING BALL	31	2	100654	PIN
14	1	100655	UNCLAMPING PIN	32	1	100218	UNCLAMPING KEY
15	1	100641	UNCLAMPING SPRING	33	1	100707	BEARING 6003 ZZ
17	1	100661	ALUMINIUM LOCK NUT	34	1	100184	CAPACITOR 8μF
17		100909	FIBRE LOCK NUT	35	1	100755	UNCLAMPING CAP
19	1	100712	STAINLESS STEEL TUBE 300mm STROKE	36	1	100748	CABLE PRESS
19	1	100713	STAINLESS STEEL TUBE 400mm STROKE	37+40	1	100716 + 100998	ENDLESS SCREW 40 + release shaft
19	1	100738	STAINLESS STEEL TUBE 600mm STROKE	37+40	1	100717 + 100998	ENDLESS SCREW 50 + release shaft
20	1	100714	BLACK ALUMINIUM TUBE 300mm STROKE	37+40	1	100740 + 100998	ENDLESS SCREW 70 + release shaft
20	1	100762	GREY ALUMINIUM TUBE 300mm STROKE	38	4	100313	CAP SCREW M5X16



ANALOGUE CONTROL UNIT FOR SWING FLEXY2 220V

IMPORTANT REMARKS.

The producer has the right to modify the product without previous notice; it also declines any responsibility to damage or injury to people, animals or things caused by improper use or wrong installation.



Please read this instruction manual very carefully before installing and programming your control unit.



This instruction manual is only for qualified technicians, who specialize in installations and automations.



The contents of this instruction manual do not concern the end user.



Every programming and/or every maintenance service should be done only by qualified technicians.



AUTOMATION MUST BE IMPLEMENTED IN COMPLIANCE WITH THE EUROPEAN RULES IN FORCE:

EN 60204-1 (machinery safety, electrical equipment of machines, part 1: general rules)

EN 12445 (Safe use of automated locking devices, test methods)

EN 12453 (Safe use of automated locking devices, requirements)



The installer must provide for a device (ex. Magnetothermal switch) ensuring the omnipolar sectioning of the equipment from the power supply. The standards require a separation of the contacts at least 3,0 mm in each pole (EN 60335-1).



After making connections on the terminal board, use one hose clamp to fix dangerous voltage wires near the terminal board and another hose clamp to fix safety low voltage wires used for accessories connection; this way, in case of accidental detachment of a conducting wire, dangerous voltage parts will not come into contact with safety low voltage ones.



The plastic case has an IP55 insulation; to connect flexible or rigid pipes fittings having the same insulation level.



Installation requires mechanical and electrical skills, therefore it shall be carried out by qualified personnel only, who can issue the Compliance Certificate concerning the whole installation (EEC Machine Directive 89/332, Annex IIA).



The automated vehicular gates shall comply with the following rules: EN 12453, EN 12445, EN 12978 as well as any local rule in force.



Also the automation upstream electric system shall comply with the laws and rules in force and be carried out workmanlike.



The door thrust force adjustment shall be measured by means of a proper tool and adjusted according to the max. limits, which EN 12453 allows.



Connect the earthing lead of the motors to the electricity grid earth system.



Observe all necessary precautions (e.g. anti-static bracelet for handling parts sensitive to electrostatic discharges).

CONFORMITY DECLARATION

The producer hereby declare that the control board conform to the essential requirements established in the following directives:

- 89/336/CEE (EMC Directive in accordance with standards EN 61000-6-2, EN 61000-6-3 + EN 50336)
- 2006/95/CEE (Low voltage Directive in accordance with standards EN 60335-1 + EN 60335-2-103)
- 99/05/CEE (Radio Directive in accordance with standard EN 301 489-3)

DESCRIPTION OF THE CONTROL UNIT

The FLEXY2 control unit is an innovative GR product guaranteeing safety and reliability for swing gate automation.

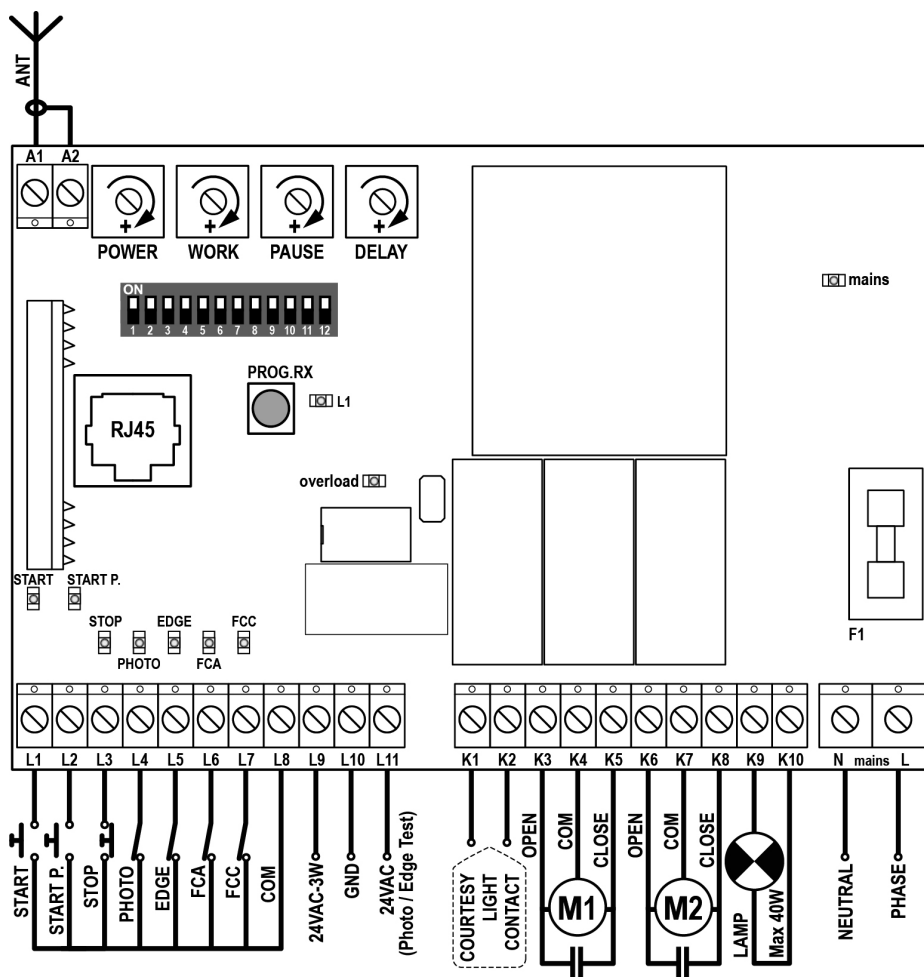
The FLEXY2 design has been aimed at creating a product which adapts to suit all needs, thus obtaining an extremely versatile control unit satisfying all the necessary requirements for a functional and efficient installation.

- 230V - 50Hz or 120V - 60Hz power supplies, depending on the model, for 2 single phase motors (700W global).
- Input for keyswitch or push-button.
- Input for safety photocell.
- Input for safety edge, capable of handling standard edges with switch normally closed and conductive rubber edges with nominal resistance of 8.2 kOhms.
- Inputs for open and close limit switches
- Pre-opening safety device testing.
- Dip-switch programmable operational logic.
- Adjustment of motor power and operation time by means of a trimmer.
- Fitted 433.92 MHz radio receiver
- Possibility of saving 240 Personal Pass transmitters (433.92 MHz)
- LED monitoring of inputs.
- Courtesy light output.
- IP55 casing.

TECHNICAL SPECIFICATIONS	230V	120V
Power supply	230V / 50Hz	120V / 60Hz
Max. motors load	700W	
Max. accessories load 24V	3,0W	
Working temp.	-20 / +60 C°	
Protection fuse	F1 = 5A delayed	F1 = 8A delayed
Dimensions	170x185x70 mm	
Weight	800 g	
Protection	IP55	

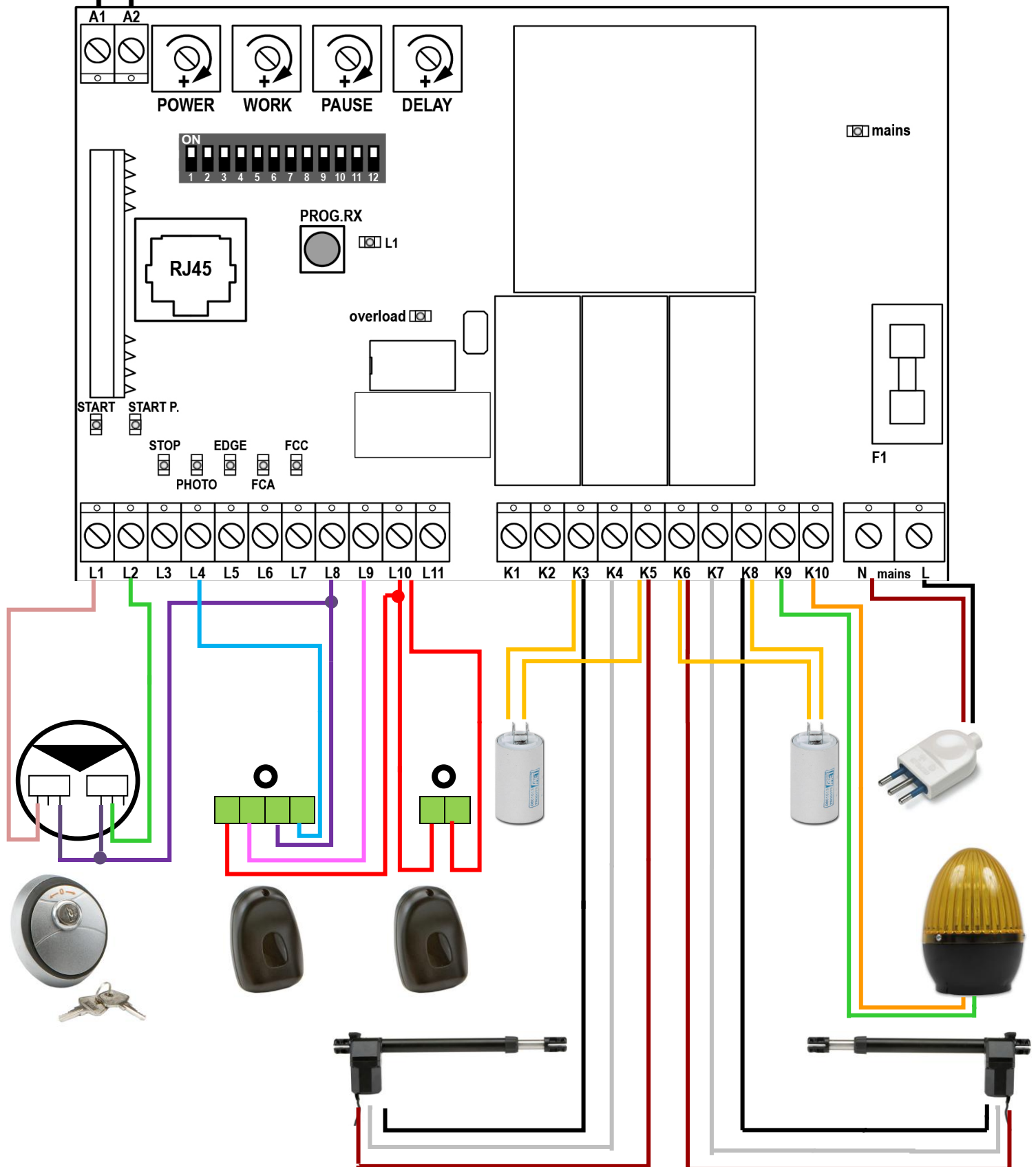
ELECTRICAL CONNECTION TABLE

PLEASE NOTE:
if not used, the normally
closed inputs
(STOP, PHOTO, EDGE)
must be jumpered with the
commands common line
COM (-)



A1	Antenna
A2	Antenna shield
L1	Opening command for a standard connection device with switch normally open.
L2	Pedestrian opening command for a standard connection device with switch normally open.
L3	STOP command N.C. switch
L4	Photocell N.C. switch
L5	Edge Switch N.C. or resistive rubber edge
L6	Limit switch open N.C. switch
L7	Limit switch closed N.C. switch
L8	Commands common (-) line
L9 – L10	24Vac power output for photocells and other accessories
L10 – L11	Power supply for functional test TX photocell and optical edge.

K1 – K2	Courtesy light timer activation switch		
		INWARD OPENING	OUTWARD OPENING
K3	Motor 1 open	BLACK	BROWN
K4	Motor 1 common	GREY	GREY
K5	Motor 1 close	BROWN	BLACK
K6	Motor 2 open	BROWN	BLACK
K7	Motor 2 common	GREY	GREY
K8	Motor 2 closed	BLACK	BROWN
K9 – K10	230V – 40W / 120V 40W blinker		
N	230V / 120V power supply neutral		
L	230V / 120V power supply phase		



INSTALLATION

Installation of the control unit, the safety devices and accessories must be performed with the power supply disconnected.

POWER SUPPLY

The control unit must be powered by means of a 230 V - 50 Hz or 120 V - 60 Hz power line, depending on the model, protected by a differential magnetothermal switch in compliance with legal regulations.

Connect the power cables to the control unit **L** and **N** terminals.

MOTORS

The control unit can control one or two asynchronous AC motors. If the control unit is used to control only one motor, then this must be connected to the terminals relating to motor 1.

Connect the cables for motor 1 as follows:

INWARD OPENING		
OPENING MOTOR 1	K3	BLACK
COMMON MOTOR 1	K4	GREY
CLOSING MOTOR 1	K5	BROWN
CAPACITOR	K3 – K5	
OPENING MOTOR 2	K6	BROWN
COMMON MOTOR 2	K7	GREY
CLOSING MOTOR 2	K8	BLACK
CAPACITOR	K6 – K8	
OUTWARD OPENING		
OPENING MOTOR 1	K3	BROWN
COMMON MOTOR 1	K4	GREY
CLOSING MOTOR 1	K5	BLACK
CAPACITOR	K3 – K5	
OPENING MOTOR 2	K6	BLACK
COMMON MOTOR 2	K7	GREY
CLOSING MOTOR 2	K8	BROWN
CAPACITOR	K6 – K8	

SAFETY EDGES

The control unit has an input for controlling safety edges; this input is capable of controlling standard edges with switch normally closed and conductive rubber edges with nominal resistance of 8.2 kOhms.

Edges can be used with two settings:

1. Edge always active:

Intervention of the edge during opening or closing causes inversion of the direction of movement in order to free the body that caused the edge to intervene.

The gate stops after approx. 3 seconds.

2. Edge NOT active during opening:

Intervention of the edge during opening is ignored. Intervention of the edge during closing causes the gate to re-open completely. Independently of the settings selected, any subsequent automatic re-closure will be cancelled.

Standard edge with switch normally closed: connect the edge cables between terminals **L5** and **L8** on the control unit.

In order to satisfy the requirements of standard EN12978, it is necessary to install safety edges with a control unit which constantly monitors correct operation. If control units are used with the option of running tests by means of interrupting the power supply, connect the control unit power supply cables between terminals **L10** (GND) and **L11** (+).

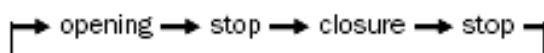
START INPUT

The START input is preset for connecting devices with the switch normally open.

Function depends on the mode of operation set by means of **dip-switch 4**.

Step mode

Subsequent Start command cause, in order:

**“Inversion” mode**

Start during opening causes closure.

Start during closure causes opening.

Start with the gate open always results in closure;

The only case where this does not occur immediately is when automatic closure is enabled and start while opening is not accepted: in this specific case, start makes the pause time count start from zero, after which the gate will be re-closed.

In both modes it is possible to disable the Start command during gate opening by means of **dip-switch 3**.

Connect the start input control device cables between terminals **L1** and **L8** on the control unit.

PEDESTRIAN START

With the gate closed, the pedestrian start command causes partial opening (approx. half way) of the gate leaf connected to motor 1. Subsequent pedestrian start commands will function according to step logic.

During a pedestrian cycle, the start command results in the complete opening of both gate leaves.

Connect the pedestrian start input control device cables between terminals **L2** and **L8** on the control unit.

PHOTOCELLS

The control unit has a 24VAC power supply for photocells with switch normally closed, and can perform an operational test before to starting the gate opening procedure. The photocell can be used with two settings:

1. Photocell always active:

Intervention of the photocell during opening or closing causes the gate to stop. When the photocell restores, the gate re-opens completely.

2. Photocell NOT active during opening:

Intervention of the photocell during opening is ignored.

Intervention of the photocell during closing causes the gate to re-open completely.

Independently of the setting selected, when the gate is paused while opening, the time count for any automatic re-closure will only start after the photocell restores.

- Connect the photocell transmitter power cables between terminals **L10** (GND) and **L11** (+) on the control unit.
- Connect the photocell receiver power cables between terminals **L10** (GND) and **L9** (+) on the control unit.
- Connect the photocell receiver output between terminals **L4** and **L8** on the control unit.

CONDUCTIVE RUBBER EDGE: connect the edge cables between terminals **L5** and **L8** on the control unit.



PLEASE NOTE: operational testing on edges is reserved for standard edges (only if equipped with suitable control units).

DO NOT enable testing if conductive rubber edges are used or standard edges used without a suitable control unit for controlling function.



NOTE: use the special interface (code 35A024) for connection of the optical bars, de-activating the operational test on the bars.

COURTESY LIGHT

This output has a normally-open clean contact relay which closes for approx. 1 second at the start of an opening phase. This switch may be used to activate a courtesy light timer (max. load: 230V - 4 A).

The switch is on terminals **K1** and **K2**.

BLINKER

The control unit provides for the use of a 230V - 40W or 120V - 40W blinker with built-in intermittence.

Connect the cables to terminals **K9** and **K10**.

STOP

The STOP input is intended for devices with the switch normally closed. The STOP command causes the immediate stop of the gate.

A subsequent START command activates the gate in the opposite direction of movement.

If the STOP command is given during opening or pause, then there will be no subsequent automatic re-closure.

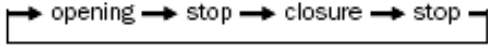
Connect the stop input control device cables between terminals **L3** and **L8** on the control unit.

EXTERNAL ANTENNA

It is recommended the external antenna be used in order to guarantee maximum radio capacity. Connect the antenna hot pole to terminal **A1** of the control unit and the braiding to terminal **A2**

PROGRAMMING THE OPERATIONAL LOGIC

It is possible for the control unit to use several different operational logic states, by simply moving the dip-switches located on the card. The functions associated with each individual dip-switch are listed below.

DIP	FUNCTION	SETTING		DESCRIPTION
1	Pre-flashing	ON	Disabled	The blinker is switched on when the motors are started
		OFF	Enabled	The blinker is switched on for 2 seconds before the motors are started
2	Automatic closure	ON	Enabled	The gate is closed automatically after the period of time set by the PAUSE trimmer
		OFF	Disabled	On completion of the opening step, the gate remains open. It is necessary to instruct closure with another START command
3	Start opening	ON	Not accepted	Any START command issued during opening is ignored
		OFF	accepted	Any START command issued during opening is accepted
4	Operational logic	ON	Inversion	Start during opening causes closure. Start during closure causes opening.
		OFF	Step	Commands subsequent to starting cause, in order: 
5	Slow down	OFF	Disabled	NOT ALLOWED FOR GR12 MOTOR
6	Start off	ON	Disabled	At the start of each opening and closing step, the motors are started at enabled maximum power.
		OFF	Enabled	
7	Anti - slip	ON	Disabled	The time used for opening or closure will always be the value set by the WORK trimmer, even if the previous operation has been interrupted before the expiry of such time.
		OFF	Enabled	When an opening (or closing) operation is interrupted before expiry of the set time (for example due to the intervention of one of the safety devices or due to a start command), the duration of the subsequent closing (or opening) operation will not be that set by the WORK trimmer, but will be equal to the time effectively elapsed, plus a short supplemental time in order to compensate for the inertia of the gate.
8	Photocell	ON	Always active	Intervention of the photocell during opening or closing causes the gate to stop. When the photocell restores, the gate re-opens completely.
		OFF	NOT active during opening	Intervention of the photocell during opening is ignored. Intervention of the photocell during closing causes the gate to re-open completely.
9	Photocell test	ON	Enabled	The control unit performs a photocell operational test before starting each opening or closing operation. If the photocells are not operating correctly, the gate does not begin to move and the light flashes for approx. 8 seconds. PLEASE NOTE: connect the photocell TX correctly
		OFF	Disabled	
10	Safety edge type	ON	Conductive rubber edge	Select this option if using conductive rubber edges with nominal resistance of 8K2.
		OFF	Standard or optical edge	Select this option if using standard edges with switch normally closed or optical edges.
11	Safety edge	ON	Always active	Intervention of the edge during opening or closing causes inversion of the direction of movement in order to free the body that caused the edge to intervene. The gate will be stopped after approx. 3 seconds.
		OFF	NOT active during opening	Intervention of the edge during opening is ignored. Intervention of the edge during closing causes the gate to be re-opened completely.
12	Safety edge test	ON	Enabled	The control unit performs an operational test on the edges before starting each opening or closing operation. If the edges are not operating correctly, the gate does not begin to move and the light flashes for approx. 8 seconds. DO NOT enable testing if conductive rubber edges are used or standard edges are used without a suitable control unit for controlling function.
		OFF	Disabled	

ADJUSTMENT OF THE POWER AND OPERATIONAL TIMES

The power and operating times may be adjusted by means of 4 trimmers located on the control unit:

POWER: motor power.

WORK: motor operating time (2 - 50 seconds).



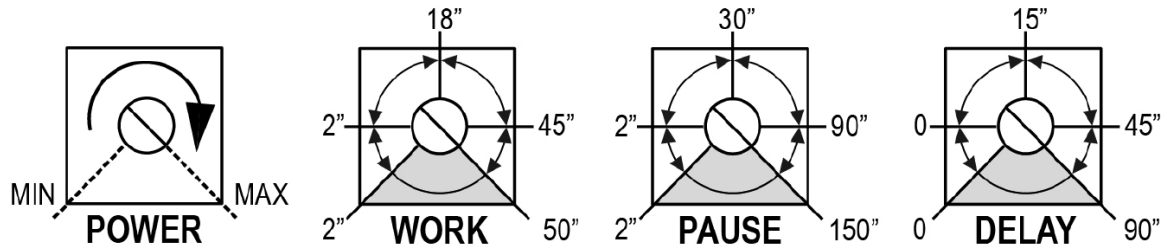
PLEASE NOTE: it is recommended that operating times be set with the slow down function disabled (DIP 5 OFF).



WARNING: the adjustment of times has to be made when the gate is still

PAUSE: pause time before automatic re-closure (2 - 150 seconds).

DELAY: time delay between the two gate leaves (0 - 90 seconds).



CONTROL UNIT INDICATORS (LEDS)

The highlighted boxes indicate the state of the LEDs when the gate is resting.

LED	ON	OFF
START	START input closed	START input open
START P.	START P. Input closed	START P. Input open
STOP	STOP input closed	STOP input open
PHOTO	PHOTO input closed	PHOTO input open
EDGE	Standard edge	
	EDGE input closed (edge not pressed)	EDGE input open (edge pressed)
	Resistive rubber edge	
	EDGE input closed (edge pressed)	EDGE input open (fault)
FCA	Opening limit switch closed	Opening limit switch open
FCC	Closing limit switch closed	Closing limit switch open
mains	Control unit powered-up	Control unit NOT powered-up
overload	Accessory power supply overload	Accessory power supply within normal operational limits